

**NAME : SHAH FAISAL**

**ROLL NO : 23I-0058**

**SECTION : BSE – 4A**

---

**OPERATING SYSTEMS LAB**

---

**BSSE SPRING 2026**

**LAB 02**

Create a file example.txt with at least 20 lines in home directory.

```
cd ~ && seq 1 20 > example.txt
```

I. Display the first 5 lines of the file.

```
head -n 5 example.txt
```

II. Display the first 10 lines of the file.

```
head -n 10 example.txt
```

III. Display the last 5 lines of the file.

```
tail -n 5 example.txt
```

IV. Display lines 6-10.

```
head -n 10 example.txt | tail -n 5
```

V. Display the first 100 bytes.

```
head -c 100 example.txt
```

VI. Display the last 100 bytes.

```
tail -c 100 example.txt
```

Create another file example2.txt with at least 30 lines.

```
seq 1 30 > example2.txt
```

VII. Display the first 3 lines of both files.

```
head -n 3 example.txt example2.txt
```

VIII. Display the last 3 lines of both files.

```
tail -n 3 example.txt example2.txt
```

IX. Search for a file named example.txt in your home directory.

```
find ~ -name "example.txt"
```

X. Find all .txt files in the /home/user/documents/ directory.

```
find /home/user/documents -name "*.txt"
```

XI. Find all files larger than 100MB in the /var/log/ directory.

```
find /var/log -type f -size +100M
```

XII. Find all files smaller than 1KB in your current directory.

```
find . -type f -size -1k
```

XIII. Find all empty files in /var/ directory.

```
find /var -type f -empty
```

XIV. Find all empty directories in your home directory.

```
find ~ -type d -empty
```

XV. Change the permissions of example.txt so that only the owner has read, write, and execute permissions, while the group and others have no permissions. Using octal number.

```
chmod 700 example.txt
```

XVI. Change the permissions of example.txt so that everyone (owner, group, others) has read and write permissions. Using octal number.

```
chmod 666 example.txt
```

XVII. Change the permissions of example.txt so that it is read-only for the owner, group, and others. Using octal number.

```
chmod 444 example.txt
```

XVIII. Change the permissions of example.txt so that others (everyone except the owner and group) have no permissions at all.

```
chmod 660 example.txt
```

XIX. Change the permissions of example.txt so that the group and others lose their write permissions but retain any other existing permissions.

```
chmod go-w example.txt
```

XX. Use symbolic notation to give the owner execute permission, the group read and execute permissions, and others only execute permission.

```
chmod u+x,g+rx,o+x example.txt
```

XXI. Change the permissions of example.txt so that everyone (owner, group, others) can execute the file, without affecting other permissions. Without using u, g and o.

```
chmod +x example.txt
```